

Weihao Lu

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🌐 <https://luweihao.github.io/>



Employment History

- 2025.2 – 2026.2 **Research Fellow**
Department of Statistics and Data Science, National National University of Singapore.
Principal Investigator: Yingcun Xia.

Education

- 2020.9 – 2024.12 **Ph.D. in Statistics**
Department of Statistics and Data Science, Tsinghua University.
Advisor: Qian Lin.
- 2015.9 – 2019.6 **B.S. in Statistics**
School of Mathematics and Statistics, Wuhan University.

Research Interest

Large-dimensional spectral algorithms, minimax risks in non-parametric regression.

Research Publications

Journal Articles

- 1 Haobo Zhang, Yicheng Li, **Weihao Lu**, and Qian Lin, “Optimal rates of kernel ridge regression under source condition in large dimensions,” *Journal of Machine Learning Research*, 2025, To appear. 🔗 URL: <https://arxiv.org/abs/2401.01270>.
- 2 Haobo Zhang, **Weihao Lu**[†], and Qian Lin, “The phase diagram of kernel interpolation in large dimensions,” *Biometrika*, vol. 112, no. 1, asae057, Nov. 2024, ISSN: 1464-3510. 🔗 DOI: [10.1093/biomet/asae057](https://doi.org/10.1093/biomet/asae057).

Conference Proceedings

- 1 **Weihao Lu**, Haobo Zhang, Yicheng Li, and Qian Lin, “On the saturation effects of spectral algorithms in large dimensions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 37, Curran Associates, Inc., 2024, pp. 7011–7059. 🔗 URL: https://proceedings.neurips.cc/paper_files/paper/2024/file/0d3496dd0cec77a999c98d35003203ca-Paper-Conference.pdf.
- 2 Haobo Zhang, Yicheng Li, **Weihao Lu**, and Qian Lin, “On the optimality of misspecified kernel ridge regression,” in *International Conference on Machine Learning (ICML)*, vol. 202, PMLR, 2023, pp. 41 331–41 353. 🔗 URL: <https://proceedings.mlr.press/v202/zhang23x.html>.

Preprints

- 1 **Weihao Lu**, Jialin Ding, Haobo Zhang, and Qian Lin, *On the pinsker bound of inner product kernel regression in large dimensions*, 2024. 🔗 URL: <https://arxiv.org/abs/2409.00915>.

- 2 **Weihao Lu**, Haobo Zhang, Yicheng Li, Manyun Xu, and Qian Lin, *Optimal rate of kernel regression in large dimensions*, 2023.  URL: <https://arxiv.org/abs/2309.04268>.

Awards

- 2023  **Top Ten Papers Award**, Ninth National Graduate Statistics Forum, Renmin University of China.
-  **Best Poster Award**, Peking-Tsinghua Joint Statistics Colloquium, Peking University.
-  **First Prize**, Seventh National Statistics Doctoral Academic Forum, Guangzhou University.
-  **Second Prize for Academic Papers**, First National Doctoral Academic Forum on Mathematics and Statistics, Liaoning University.
- 2024  **NeurIPS 2024 Scholar Award**
- 2025  **AISTATS 2025 Best Reviewer Award**
-  **NeurIPS 2025 "Reviewer : Top Reviewer" Award**

Teaching Assistants

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|---|---|
| 2020 Fall and 2021 Fall |  Elementary Probability Theory. |
| 2021 Spring |  Reliability Data and Survival Analysis. |
| 2022 Spring, 2023 Spring, and 2024 Spring |  Mathematical Statistics. |

Academic Services

-  Conference Reviewer NeurIPS, 2025
-  Conference Reviewer AISTATS, 2025, 2026